



4695 - DARK-SKY: Constrain Zodiacal Light & diffuse Extragalactic Background Light from Archival JWST images

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Most photons in JWST's near-IR images come from within 5 AU: Zodiacal Light (ZL) as seen from L2. Smaller components come from JWST straylight (SL), Diffuse Galactic Light (DGL), Integrated Galaxy Light (IGL), diffuse Extragalactic Background Light (EBL), and thermal radiation. JWST serves as absolute IR photometer on long timescales, and so can uniquely constrain these components. Archival Legacy Calibration project "DARK-SKY" will analyze the sky-surface brightness (sky-SB) observed between discrete objects detected in public images from ~100 independent sight-lines using 8 broad- and 12 medium-band NIRCcam+NIRISS filters. Our calibration and science goals are:

1) Several significant calibration improvements: high-fidelity scene-dependent 1/f-noise and wisp removal, and chip-to-chip zeropoint refinements to <2%, providing a Legacy Calibration dataset to the community.

2) Measure the 0.9-5 micron object-free sky-SB in these filters across the sky within <2-4% including full error propagation, with NIRISS as check

3) Model the sky-SB for the observing date and the 3D position of JWST in L2 as the sum of: a) ZL using updated models of solar system dust-scattering; b) SL models from the IR sky reflecting off dust on JWST's primary; c) the IGL as a best fit to the galaxy counts; and d) the DGL level derived for each direction.

4) Set competitive limits to, or detect, any remaining Diffuse Light (including diffuse EBL) from the best-fit difference between 2) and 3) across the sky. We expect our 2.7-5.0 micron Diffuse Light values to be better than 4-8 nW/m²/sr, so DARK-SKY can constrain the SED and source of this mysterious excess Diffuse Light.

OBSERVING DESCRIPTION

see attached PDF